Microphone Selection Cheat Sheet

Broadcast field quick reference | Match the pickup pattern and mic type to the source, space, and movement.

> **Fast decision:** Place the mic close to the speaker, choose the pattern that fits the space, monitor on closed-back headphones, and protect the mic from wind.

Polar patterns: what they hear

Pattern	Pickup behavior	Good fit
Omnidirectional	Sound from all directions; natural, even response.	Quiet rooms, small lavaliers, roundtable ambience. Picks up room noise.
Cardioid	Front-focused; rejects some sound from the rear.	Handheld interviews and voice in a moderately controlled space.
Super/hypercardioid	Narrower front pickup with a small rear pickup lobe.	Isolating a nearby voice; aim carefully and watch rear noise.
Shotgun / lobar	Interference tube increases directionality; off-axis tone can change.	Boomed dialogue outdoors or on set. Not a cure for echo indoors.
Figure-eight	Front and rear pickup; rejects the sides.	Two-person face-to-face interview or controlled studio use.

Pattern names describe directionality, not transducer type. A shotgun is a microphone design, not a guarantee of perfect isolation.

Transducers: how sound becomes signal

Type	Strengths / cautions	Common broadcast use
Dynamic	Rugged; usually needs no phantom power; less sensitive to quiet detail.	Handheld street interviews, loud sources, fast field work.
Condenser / electret	Sensitive, detailed; needs power (phantom or battery, depending on model).	Shotgun dialogue, studio voice, compact lavaliers.
Ribbon	Smooth sound; handle and power only as manufacturer permits.	Controlled studio voice or instruments; protect from wind and handling.

Quick scenario match

Situation	Starting choice
Noisy sidewalk interview	Close handheld cardioid dynamic + windscreen
Dialogue, mic kept out of frame	Boomed shotgun; indoors, test for reflections and consider a hypercardioid
Moving presenter, hands-free	Lavalier; secure cable and reduce clothing rub
Two voices facing each other	Two close mics, or figure-eight in a controlled room

Before recording: confirm placement, check batteries/power and connections, listen for room noise, set safe gain, record a test, and capture room tone. Manufacturer specifications vary; verify power and handling requirements for the specific microphone.